HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and developed **Python** feature pipelines on **AWS Lambda** and **S3** to power downstream ML scoring, processing approximately **3,500 events/sec** during peak hours and replacing a brittle scheduled-batch job.
- Engineered and trained a **gradient-boosted classifier** (scikit-learn) on a **~280k-row** labelled transactions dataset to flag suspicious activity, achieving **~0.86 AUC** on held-out data.
- Collaborated with enterprise customers as the primary technical contact during onboarding, gathering feature-pipeline requirements directly with their **data science teams** and translating them into **Python** implementations.
- Built secure **RESTful APIs** using **FastAPI** to expose AI capabilities for downstream products, incorporating input validation and p95 latency below **180ms**.
- Containerised feature-pipeline services with **Docker** and deployed them to an **AWS EKS Kubernetes** cluster, configuring Helm charts and basic resource limits for predictable rollouts.
- Championed **Responsible AI** principles by integrating data privacy and security considerations throughout the development lifecycle for AI solutions.
- Mentored two junior engineers on best practices for **Python** development, code reviews, and CI/CD integration within an agile framework.

SKILLS

AI/ML – Machine learning, Deep learning, Data science, Data modelling, Large language models, Natural language processing, Prompt engineering, Context engineering, Retrieval-Augmented Generation (RAG), Generative AI, Agentic solutions, Responsible AI

Programming – Python, SQL, Java, Bash

Cloud Platforms – AWS SageMaker, Azure Machine Learning, Google Vertex AI, AWS (EC2, S3, Lambda, CloudWatch)

Data Technologies – Vector databases, Knowledge graphs, Relational databases, Document databases, SQLAlchemy, PostgreSQL, MySQL, Oracle

MLOps & Deployment – MLOps, Model lifecycle management, Production monitoring, API development, Docker, Kubernetes, CI/CD, Git, Testing, Code reviews

Frameworks & Libraries – FastAPI, LangChain, scikit-learn, pandas, NumPy

PROJECTS

Customer Onboarding for ML Feature Store

- Worked directly with two pilot enterprise customers to design and ship a guided onboarding flow for an internal ML feature store, reducing time-to-first-feature from ~6 weeks to 9 days.
- Built a **Python CLI** to validate customer feature definitions against the platform schema before submission, catching ~70% of schema errors at write time instead of at training time.
- Documented common integration patterns and ran two follow-up workshops with customer data teams to upskill their internal engineers on the ML platform.

Fraud Detection Model on Transaction Data

- Engineered a feature pipeline in **Python** and **SQL** to extract behavioural signals from a transaction log of roughly **1.1M records**.
- Trained a **LightGBM classifier** on the resulting features, tuning class weights to handle the 1:140 fraud imbalance and reaching **~0.91 AUC / 0.46 F1** on a held-out month.
- Packaged the model behind a **FastAPI** inference endpoint with input validation, achieving p95 latency around **180ms** and basic feature-drift logging.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)